

Significant Figures and Measurement

Counting significant figures, scientific notation, rounding and calculations

Name _____ Class _____ Date _____

RULES

Leading zeros: not significant · Zeros between digits: significant · Trailing zeros after a decimal point: significant

× and ÷: fewest significant figures · + and -: fewest decimal places

Exact numbers do not limit the answer. Round only the final answer.

Give every answer with its unit.

A. Understanding

1. A ruler is marked in millimetres. Explain why a length measured with it can be recorded as 4.37 cm but not as 4.372 cm.

2. Explain the difference between accuracy and precision. Can a set of measurements be precise but not accurate? Give an example.

B. Counting significant figures

3. How many significant figures are in each? (a) 0.0050 m (b) 30.70 g (c) $6.022 \times 10^{23} \text{ mol}^{-1}$ (d) 0.10200 L (e) 1200 mL

C. Scientific notation and rounding

4. Write in scientific notation, keeping the significant figures: (a) 0.000 350 m (b) 45 600 g (to 3 significant figures) (c) 0.0207 L

5. Round each to three significant figures: (a) 2.4562 g (b) 0.0012349 m (c) 18 750 s (d) 9.9951 mL

D. Calculations

6. Calculate and give each answer to the correct number of significant figures: (a) $3.25 \text{ cm} \times 2.1 \text{ cm}$ (b) $100.0 \text{ g} \div 3.0 \text{ mL}$ (c) $8.315 \text{ J}/(\text{mol}\cdot\text{K}) \times 298 \text{ K}$

7. Calculate to the correct precision: (a) $12.0 \text{ g} + 0.335 \text{ g} + 4.5 \text{ g}$ (b) $6.38 \text{ cm} - 2.1 \text{ cm}$

E. Challenge

8. An aluminium block measures $2.00 \text{ cm} \times 3.00 \text{ cm} \times 4.00 \text{ cm}$ and has a mass of 64.8 g. Calculate its volume and its density, each to the correct number of significant figures.

9. Three measurements of a mass are 2.52 g, 2.48 g and 2.51 g. Calculate the mean and report it to the correct precision.